CONTACT DETAILS OF THE BODY SUBMITTING THE QUALIFICATION FILE

Name and address of submitting body:

Electronics Sector Skills Council of India(ESSCI)

Head Office: 155, 2nd Floor, ESC House

Okhla Industrial Area - Phase 3, New Delhi – 110020

Tel: +91 -84477-38-501

Name and contact details of individual dealing with the submission

Name: Dr. Abhilasha Gaur

Position in the organization: COO

Address if different from above: Same as above

Tel number(s):

E-mail address: ceo@essc-india.org

List of documents submitted in support of the Qualifications File

1. Model Curriculum

Model Curriculum to be added which will include the following:

- **Indicative list of tools/equipment to conduct the training**
- **Trainers qualification**
- **Lesson Plan**
- **Distribution of training duration into theory/practical/OJT component**

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SUMMARY

1	**Qualification Title:** Electrician
2	**Qualification Code, if any:** ELE/Q5804
3	**NCO code and occupation:** NCO-2015/7411.0100, Assembly and System Integration
4	**Nature and purpose of the qualification (Please specify whether qualification is short term or long term):** The individual at work Planning, Design & Installation modules and sub parts together that form the electrical & electronic system of the product.
5	**Body/bodies which will award the qualification:** Electronics Sector Skills Council of India
6	**Body which will accredit providers to offer courses leading to the qualification:** Electronics Sector Skills Council of India
7	**Whether accreditation/affiliation norms are already in place or not, if applicable (if yes, attach a copy):** Yes
8	**Occupation(s) to which the qualification gives access:** Assembly And Integration
9	**Job description of the occupation**: The individual at work integrates together modules and sub parts that form the electronic system of the product.
10	**Licensing requirements:** N/A
11	**Statutory and Regulatory requirement of the relevant sector (documentary evidence to be provided):** N/A
12	**Level of the qualification in the NSQF:** Level 4
13	**Anticipated volume of training/learning required to complete the qualification:** 540 hours (Theory – 90 hours, Practical – 210 hours, OJT – 180 hours, ES – 60 hours)
14	**Indicative list of training tools required to deliver this qualification:** Electronic components such as the resistor, ICs, capacitor, transistor, diode, PCB and so on Desktop, OS software (Linux, Windows), different types of circuits, Electrical sub system of the final products with remote Screw Drivers, Spanners, Drill Machine, Multi-meter, Circuit Tester, Scissors, Pliers Pencil Electrical tape, piano wire, Wall Mount Kit Antenna, STB Measuring Tape, Hammer, Crimping Tools, Cutter/ knife, Digital IC tester with manual/Batch CRO Soldering Tool Kit, SMD Soldering Tools Manual Guide, Trainer Kit, Tested board, bare board and assembled board etc.
15	**Entry requirements and/or recommendations and minimum age:** 18 Years

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	8th grade pass + NTC (2 years after 8th) with 2 years of NAC/relevant experience OR 10th grade pass with 2 years NTC/NAC/relevant experience OR 12th grade pass OR Certificate (NSQF Level-3 in the Electrical Technician) with 2 years of relevant experience		
16	**Progression from the qualification (Please show Professional and academic progression):** Head of Electrician		
17	**Arrangements for the Recognition of Prior learning (RPL):** RPL will be based on the same approved Qualification Pack and Assessment Criteria mentioned in the Qualification Pack by Electronics Sector Skills Council of India (ESSCI)		
18	**International comparability where known (research evidence to be provided):** Yet to establish		
19	**Date of planned review of the qualification:** 17/11/2025		
20	**Credits Assigned to this Qualification, Subject to Assessment** *(as per National Credit Framework (NCrF))*: 18		
21	**Formal structure of the qualification** **Mandatory components**		
	Title of component and identification code/NOSs/Learning outcomes	**Estimated size (learning hours)**	**Level**
(i)	Bridge Module	04	4
(ii)	Planning, Design & Installation of Electrical and Electronic Sub System (ELE/N5806)	228	4
(iii)	Testing, Commissioning, Maintenance, Fault finding & Repair (ELE/N5805)	208	4
(iv)	Employability Skills (DGT/VSQ/N0102)	60	4
(v)	Apply health and safety practices at the workplace (ELE/N1002)	40	4
	Total	**540**	

SECTION 1
ASSESSMENT

22	**Body/Bodies which will carry out assessment:** Electronics Sector Skills Council of India
23	**How will RPL assessment be managed and who will carry it out?** Give details of how RPL assessment for the qualification will be carried out and quality assured. The RPL assessment will be carried out through pre assessment, identifying the skills gaps, provide bridge training to cover the competency gap and then conduct final assessment of the candidates.
24	**Describe the overall assessment strategy and specific arrangements which have been put in place to ensure that assessment is always valid, reliable and fair and show that these are in line with the requirements of the NSQF.** Assessment is done through third parties who are affiliated to ESSCI as Assessment Body. Assessors are trained & certified by ESSCI through Training of Assessors program. The assessment involves two processes. The first process is gathering the evidence of the competency of individuals. The second part of the assessment process is the judgement as to whether a person is competent or not. The assessment plan contains the following information: • What will be assessed, i.e. the competency based on each NOS • How assessment will occur i.e. methods of assessment • When the assessment will occur • Where the assessment will take place i.e. context of the assessment (workplace/simulation) • The criteria for decision making i.e. those aspects that will guide judgements and Where appropriate, any supplementary criteria used to make a judgement on the level of performance. The assessment is conducted through theory, viva voce and practical.

Please attach most relevant and recent documents giving further information about assessment and/or RPL.
Give the titles and other relevant details of the document(s) here. Include page references showing where to find the relevant information.

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ASSESSMENT EVIDENCE

Complete a grid for each component as listed in "Formal structure of the qualification" in the Summary.

NOTE: this grid can be replaced by any part of the qualification documentation which shows the same information – i.e. Learning Outcomes to be assessed, assessment criteria and the means of assessment.

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25. Assessment evidences

Title of Component: Electrician

CRITERIA FOR ASSESSMENT OF TRAINEES

Job Role Electrician

Qualification Pack ELE/Q5804

Sector Skill Council Electronics Sector Skills Council of India

Guidelines for Assessment

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

ELE/N5806: Planning, Design & Installation of electrical sub system				
Assessment Criteria for Outcomes	**Theory Marks**	**Practical Marks**	**Project Marks**	**Viva Marks**
Understand work requirements from the manager	**20**	**30**	**-**	**-**
PC1. communicate with the supervisor in order to understand the day's production requirement and plan work	10	15	-	-
PC2. apply approved drawings, job instructions or work manuals	10	15	-	-
Read, interpret and revise drawings and documentation	**20**	**30**	**-**	**-**
PC3. collect consumables such as wires and connectors need for the assembly process	2	4	-	-
PC4. collect the work manual/job instruction from the manager	2	4	-	-
PC5. visually inspect the boards and components received for any physical damage, any loose or wrong connections	2	4	-	-
PC6. follow the standard operating procedure to assemble the electrical sub system	2	4	-	-
PC7. interpret accurately drawings, wiring and job specifications/instructions	4	4	-	-
PC8. underline any errors in previous step of the assembly process identified	2	4	-	-
PC9. report defective or inadequate number of components	4	4	-	-
PC10. report about inadequate quantity of consumables such as wires, connectors, screws, nuts, etc.	2	2	-	-
NOS Total	**40**	**60**	**-**	**-**

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ELE/N5805: Testing, Commissioning, Maintenance, Fault Finding & Repair

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC1. Test installations before energizing to ensure personal and electrical safety to include: insulation resistance and earth continuity tests, correct polarity and complete a visual inspection	4	5	-	-
PC2. Test installations when energized by checking complete function on all equipment installed to ensure correct operation of new installation as per instructions, for example correct voltage, phase rotation and correct functioning of protection devices	4	5	-	-
PC3. Set-up equipment to include: selecting and using the appropriate software for programming programmable relays, bus-system; creating necessary settings on devices such as timers and overload relays; downloading and importing applications required and programming bus-system, for example KNX, DALI, Modbus and IP or IT based systems	**6**	**10**	**-**	**-**
PC4. Set the installation to fully functioning and ensure customer can operate.	4	6	-	-
Maintenance, fault finding and repair			-	-
PC5. Adapt to changing circumstances	2	4	-	-
PC6. Troubleshoot electrical installation and identify faults including: short and open circuits, incorrect polarity, insulation resistance and earth continuity faults, incorrect settings on equipment and incorrect program on programmable devices	2	4	-	-

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PC7. Diagnose electrical installation and identify problems including; bad connections, incorrect wiring, high loop impedance and equipment failure	2	4	-	-
PC8. Verify that an existing electrical installation still meets current standards.	4	6	-	-
PC9. Use, test and calibrate measuring equipment including: insulation resistance, continuity and installation testers, multi, clamp and network cable testers	4	6	-	-
PC10. Repair and replace faulty components in electrical installations	4	6	-	-
PC11. Rewire and repair faulty installations	4	4	-	-
NOS TOTAL	**40**	**60**		

DGT/VSQ/N0102: Employability Skills (60 Hours)				
Assessment Criteria for Outcomes	**Theory Marks**	**Practical Marks**	**Project Marks**	**Viva Marks**
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobsin various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. andpersonal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st CenturySkills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mailsetc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

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PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriatelywith all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

ELE/N1002: Apply health and safety practices at the workplace				
Assessment Criteria for Outcomes	**Theory Marks**	**Practical Marks**	**Project Marks**	**Viva Marks**
Deal with workplace hazards	**20**	**31**	**-**	**-**
PC1. identify job-site hazards and possible causes of accident in the workplace	2	3	-	-
PC2. perform work complying to organizational safe working practices and observing hazard signs displayed on containers, equipment and in various work areas such as inside buildings, in open areas and public spaces, etc.	3	4	-	-
PC3. use appropriate personal protective equipment (PPE) for specific tasks and work conditions, contaminant (concentration w.r.t air) requirements and severity of hazard while conforming to the Indian/International standards	3	4	-	-
PC4. follow standard safety procedures while handling tool, equipment, hazardous substances and while working in hazardous environments	3	4	-	-
PC5. dispose electronic waste (such as toxins; metals such as lead, cadmium, barium; flame retardant plastics, welding slag etc.) as per industryapproved techniques	2	4	-	-
PC6. avoid damage of components due to negligence in electrostatic discharge (ESD) procedures	2	3	-	-
PC7. locate general health and safety equipment inthe workplace such as fire extinguishers; first aid equipment; safety instruments, clothing and installations (fire exits, exhaust fans)	2	3	-	-
PC8. maintain appropriate posture while handling heavy objects	1	3	-	-
PC9. apply good housekeeping practices at all times	2	3	-	-
Apply fire safety practices	**4**	**9**	**-**	**-**
PC10. take preventive measures to prevent firehazards	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. • use appropriate fire extinguishers for different types of fires • Types of fires: Class A: e.g. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: e.g. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no l	1	3	-	-
PC12. exhibit rescue and first-aid techniques in caseof fire or electrocution	1	3	-	-
Follow emergencies, rescue and first-aid procedures	6	13	-	-
PC13. administer appropriate first aid to victims incase of bleeding, burns, choking, electric shock, poisoning etc.	1	3	-	-
PC14. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock,	1	2	-	-
PC15. participate regularly in emergency procedures such as raising alarm, safe/efficient, evacuation, correct means of taking shelter and escaping, correct assembly point, roll call, correct return to work	2	4	-	-
PC16. use correct method to move injured peopleand others during an emergency	2	4	-	-
Effective waste management/recycling practices	5	12	-	-
PC17. identify recyclable and non-recyclable, and hazardous waste generated	1	3	-	-
PC18. segregate waste into different categories	1	2	-	-
PC19. ensure disposal of non-recyclable waste appropriately	1	2	-	-
PC20. deposit non-recyclable and reusable materialat identified location	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	**35**	**65**	-	-

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Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
Provided in the above section	

Means of assessment 1

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below.)
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on these criteria.

Means of assessment 2

Add boxes as required.

Pass/Fail

1. To pass the Qualification Pack, every trainee should score a minimum of 70% in every NOS.
2. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

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SECTION 2

25. EVIDENCE OF LEVEL

Title/Name of qualification/component: Electrician			**Level: 4**
NSQF Domain	**Key requirements of the job role**	**How the job role relates to the NSQF level descriptors**	**NSQF Level**
Process	**Demands a wide range of specialised technical skill, clarity of knowledge and practice in broad range of activity involving standard and non-standard practices.** • Integrating the electrical sub system • Reporting problems to the superior • Interacting effectively with colleagues and superiors • Applying health and safety practices at the workplace.	The individual at work Planning, Design & Installation modules and sub parts together that form the electrical & electronic system of the product.	4
Professional knowledge	**Factual and theoretical knowledge in broad contexts within a field of work or study.** • Knowledge of the electrical sub system • Knowledge of company's policy on turnaround time, working hours	The individual at work Planning, Design & Installation modules and sub parts together that form the electronic system of the product. Hence Level 4	4

Title/Name of qualification/component: Electrician			**Level: 4**
NSQF Domain	**Key requirements of the job role**	**How the job role relates to the NSQF level descriptors**	**NSQF Level**
	• Achieving productivity, quality and safety standards as per company's policy		
Professional skill	**A range of cognitive and practical skills required to generate solutions to specific problems in a field of work or study.** • Communicate with the team to understand the work requirement • Complete the documentation of assembly	An Electrician must be able to communicate effectively to the supervisor about any problems faced and updation of work. Hence Level 4	4
Core skill	Reasonably good in: • Assembling of electrical and electronic sub system • Achieving productivity, quality and safety standards as per company's policy	An Electrician has to assemble the electrical and electronic sub system and should be able to point out errors to the supervisor in order to achieve productivity. Hence Level 4	4
Responsibility	Responsibility of completing the work assigned and reporting the same as per standards. • Understand the job role and follow the organisational policy • Follow safety regulations at work place	An Electrician should understand the job role and follow safety regulations at the work place along with an effective interaction. Hence Level 4	4

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Title/Name of qualification/component: Electrician			**Level: 4**
NSQF Domain	**Key requirements of the job role**	**How the job role relates to the NSQF level descriptors**	**NSQF Level**
	• Work and interact effectively with colleagues and superiors		

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SECTION 3

26. EVIDENCE OF NEED

What evidence is there that the qualification is needed? What is the estimated uptake of this qualification and what is the basis of this estimate?

Basis	In case of SSC	In case of other Awarding Bodies (Institutes under Central Ministries and states departments)
Need of the qualification The Indian electronics industry is one of the largest and fastest-growing industries in the world. The market growth of Industrial Automation is expected to be at a CAGR of 11% .	The SSC would undertake market study and would enclose demand forecast for the proposed job role both on short-term and long-term basis to substantiate the requirement of the Qualification proposed. The SSC can produce the data from primary or authorized secondary sources as well.	The Submitting Body would produce any reputable and reliable research reports, such as labour market information reports; occupational mapping or similar research carried out by Ministry/State/Any other authentic source forecasting the demand for the proposed qualification
Industry Relevance We are in the process of taking industry validation.	The SSC would undertake validation of the job roles with actual end-user industry where such employment is going to be generated and absorbed instead of generic validation of industry. The SSC would submit the endorsements from users/intended users of the qualification clearly supporting or otherwise the need for trained people against specific job role. (*The*	The Submitting Body would submit the list of industry participation while preparation of the curriculum/ course content of the qualifications. These could include minutes of the meeting/ reports of these consultations

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		industry validation format to be used)	
	Usage of the qualification: This Qualification Pack will be used across Electronic industry which is organised as well as unorganised	The SSC would submit details of the employment generated (wherever applicable) and realised by virtue of training in the Qualifications of the sector earlier submitted for NSQF alignment. In case of unorganized sector, case studies or evidences may be given	The submitting body would submit the details of trained and placed data in the proposed qualification (if an existing qualification is being proposed for NSQF alignment) Information about the success of the qualification should be given (e.g. uptake figures, examples of use in recruitment and placement rates (if known) should be given. However, many of the bodies that do not have placement tracking mechanism established in place would provide necessary endorsements by the state/ ministry stating that a tracking mechanism would be institutionalized and placement records shall be provided annually or later, depending on length of qualification.
	Estimated uptake	The SSC would submit the estimated uptake of the qualification and What steps were carried out to test the likely uptake of the qualification? The basis of this estimate should include data about the number of jobs or places	The Submitting Body should submit the estimated uptake by reflecting the number of the takers for this qualification for at least two years from submission of the qualification

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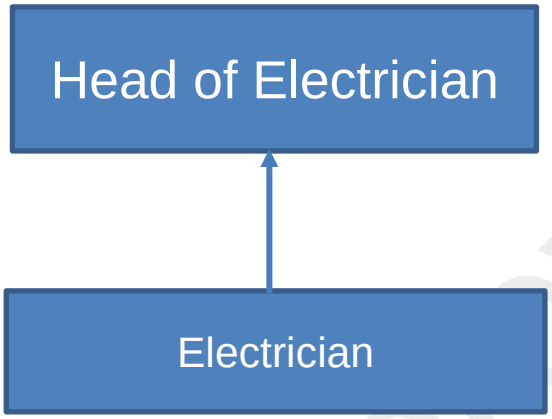
		in courses of learning which will be available to people who are awarded the qualification.		
27	**Recommendation from the concerned Line Ministry of the Government/Regulatory Body. To be supported by documentary evidences** N/A			
28	**What steps were taken to ensure that the qualification(s) does (do) not duplicate already existing or planned qualifications in the NSQF? Give justification for presenting a duplicate qualification** This is the first time that this Qualification is being made. The national qualification registers as well as the Qualification Packs with NSDC have been checked.			
29	**What arrangements are in place to monitor and review the qualification(s)? What data will be used and at what point will the qualification(s) be revised or updated? Specify the review process here** • Agencies have been appointed by the SSC to interact with training providers to gather feedback in implementation. • Monitoring of results of assessments • Employer feedback will be sought post-placement • A formal internal review is scheduled in a year			

SECTION 4
EVIDENCE OF PROGRESSION

30	**What steps have been taken in the design of this or other qualifications to ensure that there is a clear path to other qualifications in this sector?** ***Show the career map here to reflect the clear progression***

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1. Discussing the growth trajectory within each occupation after studying organisational charts of various industry players across small, medium and large-scale organizations.
2. Exploring various lateral career opportunities for the discussed qualification
3. Ensuring that there is a clear role up in terms of performance criteria qualification experience and skill requirement from lower NSQF Level to higher levels in the hierarchy.



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